

Project Report: AQI Predictor Dashboard By Maryam Zaheer

1. Project Summary

Attribute	Details
Project Title	AQI Predictor Dashboard
Developer	Maryam
Project Type	Real-time Air Quality Monitoring and Forecasting System
Technology Stack	Python, Streamlit, Machine Learning, Data Visualization
Deployment	Streamlit Cloud (Local and Production Ready)

2. Project Overview

Objective

To develop a real-time Air Quality Index (AQI) monitoring and forecasting dashboard that provides accurate predictions using machine learning and offers a user-friendly interface for visualizing air quality trends and alerts.

Problem Statement

Air pollution is a major environmental and health issue. There is a growing demand for:

- Real-time AQI monitoring
- Accurate air pollution forecasts
- City-wise comparisons
- Alerts for hazardous conditions
- User-friendly visualizations

Proposed Solution

A web-based dashboard combining real-time data, machine learning predictions, and interactive visualizations to give actionable insights into air quality.

3. System Architecture

The system follows a sequential data pipeline from data acquisition to user interaction, as described below:

Stage	Description
1. Data Source	Real-time and historical air quality data is fetched using the OpenWeatherMap API .
2. Data Processing	Data is cleaned, normalized, and transformed using Pandas and NumPy .
3. Machine Learning Models	Predictions are generated using models such as XGBoost , LightGBM , and LSTM (TensorFlow) .
4. Dashboard	The processed data and model outputs are rendered via Streamlit and visualized using Plotly .
5. User Interface	Users interact with the dashboard through an intuitive and responsive Web UI .

4. Technology Stack

Category	Tools/Frameworks
Frontend	Streamlit, Plotly, HTML5, Custom CSS (Dark Navy Theme)
Backend	Python 3.8+, Pandas, NumPy, Requests
Machine Learning	XGBoost, LightGBM, LSTM (TensorFlow/Keras), Scikit-learn, SHAP, LIME
DevOps	GitHub Actions, requirements.txt, Streamlit Cloud
Security	Environment Variables for API keys

5. Key Features

1. Real-time AQI Monitoring

- Current AQI and pollutant data: PM2.5, PM10, NO2, SO2, CO, O3
- Supports multiple cities with hourly updates from OpenWeatherMap API
- Auto-refresh functionality

2. 3-Day AQI Forecasting

- Ensemble modeling (XGBoost, LightGBM, LSTM)
- Trend visualization and model comparison
- City-specific predictions

3. Historical Data Analysis

- 30-day AQI trends with time-series visualizations
- CSV export and statistical summaries

4. ML Pipeline Integration

- Uses Hopsworks for feature store
- Automated retraining and performance monitoring
- Model explainability using SHAP and LIME

5. Hazardous AQI Alerts

- Real-time threshold monitoring
- Severity-based multi-level alert system
- Historical tracking of alerts

6. Exploratory Data Analysis (EDA)

- Correlation analysis of pollutants
- Seasonal and weekly AQI patterns
- Interactive charts and summaries

6. Implementation Details

Data Pipeline

Step	Description
Data Collection	OpenWeatherMap API for real-time & historical AQI, multi-city support
Data Processing	Feature engineering, data cleaning, normalization
Model Training	Supervised learning using ensemble and neural models

7. Model Performance

Model	R ² Score	Training Time	Remarks
XGBoost	1.0000	~2 minutes	High accuracy, fast inference
LightGBM	0.9989	~1 minute	Lightweight and efficient
LSTM	0.8727	~5 minutes	Effective temporal modeling
Random Forest	95% acc.	N/A	Used for hazard classification

8. User Interface (UI)

Dashboard Layout

Section	Contents
Header Tabs	Dashboard, Forecast, Historical, Alerts, Advanced
Sidebar	City Selector, Quick Stats, Control Options, Feature List

Visual Design

- Dark navy blue theme
- Inter font family
- Responsive layout
- Animated and interactive components

9. Technical Features

CI/CD Automation

- GitHub Actions for testing and deployment
- Automated hourly data refresh
- Model retraining integration

Feature Store

- Hopsworks-based versioning and centralized storage
- Integrated into ML training pipeline

Model Explainability

- SHAP and LIME visualizations
- Per-prediction transparency
- Supports AI regulation and trust-building

Real-time Alerts

- Probability-based severity scoring
- Historical event logging

10. System Performance

Metric	Value
Inference Time	< 1 second
Dashboard Load Time	2–3 seconds
Users Supported	50+ (Streamlit Cloud)
Forecast Accuracy	85–90%
Hazard Detection	92% precision
Data Completeness	~95%

11. Deployment & Scalability

Deployment

- Hosted on Streamlit Cloud
- Custom domain support
- Local storage for models

Scalability

- Modular, cloud-native architecture
- API integration ready
- Docker-compatible
- Future deployment on AWS, Azure, GCP supported

12. Environment Setup

Python 3.8+
Virtual Environment (venv or conda)
Install dependencies via requirements.txt
Set API keys using environment variables

13. Future Enhancements

Timeline	Planned Features
Short-Term (1–3 months)	- Integration with government & satellite data - Drift detection - Personalized alerts
Long-Term (6–12 months)	- Global city support - Health impact analytics - Developer APIs - Custom dashboards

14. Security & Compliance

Aspect	Details
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Data Security	Environment variables, HTTPS, no user data stored
Compliance	Licensed API use, AI transparency tools, aligns with AQI standards

15. Key Innovations

Technical

- Multi-model ensemble for AQI forecasting
- Real-time explainability (SHAP & LIME)
- Integrated MLOps pipeline
- City-specific model tuning

User Experience

- Elegant dark theme
- Centralized dashboard
- Probability-based alerts
- Educational components to raise awareness

16. Learning Outcomes

Technical Skills

- Full-stack development (frontend + backend + ML)
- MLOps: CI/CD, automation, monitoring
- Data engineering and feature store integration
- Streamlit Cloud deployment

Project Management

- Agile methodology
- Prioritization of features
- Model/system validation
- End-to-end documentation

